

Shadab Tabatabaeian

 Website |  Email |  225.349.6783 |  LinkedIn

Washington D.C

RESEARCH INTERESTS

My research centers on the cognitive and behavioral mechanisms underlying creative reasoning and innovation, in education. My approach is interdisciplinary, drawing on cognitive science, information theory, complex systems, machine learning, and AI fine-tuning techniques. Currently, I study how the increasing use of AI tools is reshaping the landscape of creativity in education. My goal is to develop frameworks for the responsible and effective integration of AI into teaching and learning, from K–12 through higher education.

EMPLOYMENT

2025 - Present Postdoctoral Researcher, Psychology department at **Georgetown University, D.C**, The lab for Relational Cognition, Supervisor: Dr. Adam Green.
2023 - 2025 Senior Data Scientist for a consortium of K-12 schools (**MVA, CPA, PCA**), **San Diego**
2015 - 2022 Teaching Assistant at **University of California, Merced**
2012 - 2015 Instructor of Farsi Literature and Linguistics **Roshan high-school, Tehran, Iran**

EDUCATION

2018 - 2023 Ph.D. in Cognitive and Information Sciences at **University of California, Merced**, Advisor: Dr. Tyler Marghetis, Dr. Carolyn Dicey Jennings
2018 - 2022 M.S. in Cognitive and Information Sciences at **University of California, Merced**
2015 - 2018 M.A. in Interdisciplinary Humanities at **University of California, Merced**
2010 - 2014 B.Sc. in Archaeology at **University of Tehran, Iran**
2008 - 2012 B.A. in Farsi Literature and Linguistics at **University of Tehran, Iran**

PUBLICATIONS

- Tabatabaeian, S., Green, A., Johnson, D., & Moon, K. (2026). *AI narcissism in creativity evaluation: The case of college admission essays* [Manuscript in preparation].
- Tabatabaeian, S., Ortega, A., Deluna O'bi, A., Landy, D., & Marghetis, T. (2026a). Cognitive niche construction in expert reasoning: The case of mathematicians [Under review at Cognitive Science Journal].
- Tabatabaeian, S., Ortega, A., Deluna O'bi, A., Landy, D., & Marghetis, T. (2026b). Stepping back to see the big picture: Why physical movements can inspire insight [Under review at Psychological Science].
- Tabatabaeian, S., Deluna O'bi, A., Landy, D., & Marghetis, T. (2025). An information-theoretic foreshadowing of mathematicians' sudden insights. *Proc. Natl. Acad. Sci. U.S.A.*, 122(35), e2502791122. <https://doi.org/10.1073/pnas.2502791122>
- Tabatabaeian, S., Ortega, A., Deluna O'bi, A., Landy, D., & Marghetis, T. (2024). Stepping back to see the connection: Movement during problem solving facilitates creative insight. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46(46).

- Jennings, C. D., & Tabatabaeian, S. (2023). Attention, technology, and creativity. In D. G. Burnett & J. E. H. Smith (Eds.), *Scenes of attention: Essays on mind, time, and the senses* (pp. 124–141). Columbia University Press.
- Tabatabaeian, S., Deluna O’bi, A., Landy, D., & Marghetis, T. (2023). What does the body do, when the body is doing mathematics? *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45(45).
- Tabatabaeian, S., & Jennings, C. D. (2023). Dynamic attentional mechanisms of creative cognition. *Philosophy and the Mind Sciences*, 4.
- Tabatabaeian, S., & Marghetis, T. (2023). Seeing the connection: Manipulating access to visual information facilitates creative insight. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45(45).
- Tabatabaeian, S., Deluna O’bi, A., Landy, D., & Marghetis, T. (2022). Mathematical insights as novel connections: Evidence from expert mathematicians. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44(44).
- Lobato, E. J. C., Tabatabaeian, S., Fleming, M., Sulzmann, S., & Holbrook, C. (2019). Religiosity predicts evidentiary standards. *Social Psychological and Personality Science*, 11(4), 546–551.
- Tabatabaeian, S., Holbrook, C., & Jennings, C. D. (2019). A re-examination of the interrelationships between attention, eye behavior, and creative thought. *Proceedings of the Annual Conference of the Cognitive Science Society*, 41(41).
- Tabatabaeian, S., & Jennings, C. D. (2018). Toward a neurophysiological foundation for altered states of consciousness. *Behavioral and Brain Sciences*, 41.

CONFERENCE TALKS

- **Tabatabaeian, S.** (January, 2026). Rigorous Illustrations - Their creation and evaluation for mathematical research. *Illustration as a mathematical research technique*, Henri Poincaré Institution, Paris, France.
- **Tabatabaeian, S.,** Ortega, A., Deluna O’bi, A., Landy, D., & Marghetis, T. (July, 2024). Stepping back to see the connection: Movement during problem solving facilitates creative insight. *47th Annual Meeting of The Cognitive Science Society*, Rotterdam, Netherlands. [19% acceptance rate]
- **Tabatabaeian, S.,** Deluna O’bi, A., Landy, D., & Marghetis, T. (July, 2022). Mathematical insights as novel connections: Evidence from expert mathematicians. *44th Annual Meeting of The Cognitive Science Society*, Toronto, Canada. [27% acceptance rate]
- **Tabatabaeian, S.,** Deluna O’bi, A., Landy, D., & Marghetis, T. (October, 2021). Mathematical insight within the distributed system of brain-body-blackboard. *Complexity and Cognition (CompCog) Conference*. Lyon, France.
- **Tabatabaeian, S.,** Holbrook, C., & Jennings, C. D. (October, 2020). The missing link: a new account of the link between creative cognition and altered states of consciousness. *Society for the Neuroscience of Creativity (SfNC)*. Virtual Conference.
- **Tabatabaeian, S.** (December, 2017). Effects of darkness on perceptual and cognitive processes: creativity enhancement. *116th Annual Meeting of the American Anthropological Association*. Washington, DC.

SELECTED RESEARCH PRESENTATIONS

- *An information-theoretic foreshadowing of mathematicians' sudden insights.* Computational Social Scientist Lab, Information Science Department, Cornell University, New York. November, 2026.
- *Information-theoretic early warning signals of mathematical insight.* Department of Cognitive and Information Sciences, University of California, Merced. April, 2023.
- *Seeing the connection: Manipulating access to visual information facilitates creative insight.* Language, Interaction, & Cognition Lab, University of California, Merced. March, 2023.
- *Dynamic attentional mechanisms of creative cognition.* Philosophy Lab. University of California, Merced. February, 2023.
- *Modeling mathematical creativity in its natural habitat.* Department of Cognitive and Information Sciences, University of California, Merced. February, 2022.
- *Creative performance under threat: A Bayesian analysis.* Advanced Statistics for Cognitive Science Course, Dr. Rachel Ryskin, University of California, Merced. November, 2021.
- *Modeling neural correlates of creative cognition via simulated annealing algorithm.* Complex Adaptive Systems Course, Dr. Michael Spivey, University of California, Merced. April, 2020.
- *A re-examination of the interrelationships between attention, eye behavior, and creative thought.* Department of Cognitive and Information Sciences, University of California, Merced. May, 2019.

ORGANIZED WORKSHOPS

- **Learning in the age of AI.** Organized a professional development workshop for school admins across three K–12 schools in San Diego. The session covered effective learning strategies, the role of AI in supporting learning, and hands-on activities to develop skills in prompt crafting, evaluation of AI outputs, and identifying appropriate educational use cases. (Mission Vista Academy, Cabrillo Point Academy, Pacific Coast Academy; San Diego, CA, January, 2025)
- **Intro to Data Science with R and Python.** Conducted a workshop to help Master's students in Cognitive Science with data analysis and visualization in R and Python. Included hands-on training in data manipulation, statistical analysis, and creating visualizations to effectively communicate research findings (Spring 2023).
- **Designing online behavioral experiments in Qualtrics.** Trained undergraduate research assistants to create and conduct online studies, focusing on experiment design and data collection. In this workshop, I led students in developing a hypothesis about how joint attention (for example, being able to see the instructor's cursor) could enhance the quality of online Python coding tutorials. I guided them through stimulus design, experiment implementation in Qualtrics, and later data analysis. (Marghetis Lab, Fall 2023).
- **Coding Gestures in Naturalistic Videos and Calculating Inter-Rater Reliability.** Led a workshop for undergraduate research assistants focused on coding gestures, resolving rater disagreements, and statistical methods for reliability assessment (Marghetis Lab, Spring 2022).
- **Using eye-tracking technology to capture cognitive processes.** Trained undergraduate research assistants on the utilization of eye-tracking devices in experimental research. The workshop focused on equipment setup, experimental design, data collection and analysis (Holbrook Lab, 2019).
- **Time series data analysis in R.** Led a specialized workshop for PhD students on time series data analysis, covering key concepts, methodologies, and advanced techniques for analyzing time-dependent data to uncover trends (Spring 2021).

SELECTED POSTERS

- **Tabatabaeian, S.**, Moon, K., Johnson, D., Kay, M., & Green, A. (June, 2026). AI Narcissism: The Persistence of Self-Preference Bias in LLM Evaluations of Creativity. *Society for the Neuroscience of Creativity*, Philadelphia, US.
- **Tabatabaeian, S.**, Liu, Y., Johnson, D., Patterson, J.D., & Green, A. (June, 2026). Mitigating Creative Homogenization in Human–AI Collaboration Through Real-Time Semantic Feedback. *Society for the Neuroscience of Creativity*, Philadelphia, US.
- **Tabatabaeian, S.**, Deluna O’bi, A., Landy, D., & Marghetis, T. (July, 2023). What does the body do, when the body is doing mathematics? *45th Annual Meeting of The Cognitive Science Society*, Sydney, Australia.
- **Tabatabaeian, S.**, & Marghetis, T. (July, 2023). Seeing the connection: Manipulating access to visual information facilitates creative insight. *45th Annual Meeting of The Cognitive Science Society*, Sydney, Australia.
- **Tabatabaeian, S.** (April, 2018). Potential Links between Eye-movement Trajectories and Creative Thought. *10th Annual California Cognitive Science Conference (CCSC)*, University of California, Berkeley.

PRESS COVERAGE AND MEDIA INTERVIEWS

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| 11/10/2025 | Scientists See ‘Eureka’ Moments in Mathematicians’ Chalkboard Writings, Scientific American. Link Here |
| 08/28/2025 | You Can Catch the Signs of a Eureka Moment Right Before It Explodes in Your Brain, Popular Mechanics, Science. Link Here |
| 08/24/2025 | Just before the coin falls: Researchers have identified a unique pattern that precedes “aha moments”, Haaretz, Science. Link Here |
| 08/22/2025 | Heureka: Aha-Momente Kündigen sich an, Deutschlandfunk, Interview with German national Radio. Link Here |
| 08/21/2025 | Eurêka! Une étude met en lumière le processus qui déclenche les éclairs de génie scientifiques, “Sciences, Mathématiques,” Le Nouvel Obs. Link Here |
| 08/18/2025 | Scientists show signs of ‘eureka moments’ before they happen, “News,” Times Higher Education. Link Here |

EDITORIAL AND PEER-REVIEW EXPERIENCE

- Journal reviewer for Cognitive Science (2024).
- Collaborated in editing and proofreading the following academic publication: Jennings, C. D. (2020). *The Attending Mind*. Cambridge: Cambridge University Press.
- Journal reviewer for Developmental Science (2019).

TEACHING AND TUTORING EXPERIENCE

- **Intro to Data Science with Python Workshop**, Instructor, University of California, Merced (Spring 2023).
- **Culture and Cognition**, Dr. Tyler Marghetis, Teaching Assistant, University of California, Merced (Fall 2022).
- **Graduate Student Academic Writing Tutor**, Teaching Assistant, University of California, Merced (Spring 2021).

- **Critical Reasoning**, Dr. Daniel Hicks, Teaching Assistant, University of California, Merced (Fall 2021).
- **Neuroscience of Consciousness**, Dr. Carolyn Dicey Jennings, Teaching Assistant, University of California, Merced (Spring 2020).
- **Contemporary Moral Issues in Philosophy**, Dr. David Jennings, Teaching Assistant, University of California, Merced (Spring 2019).
- **Introduction to Philosophy**, Dr. Rolf Johansson, University of California, Merced (Fall 2019).
- **Introduction to Cognitive Science**, Dr. Michael Spivey (Fall 2018 and Fall 2015), Dr. Paul Smaldino (Spring 2018), Teaching Assistant, University of California, Merced.
- **Research Methods in Cognitive Science**, Dr. Rick Dale (Spring 2017), Teaching Assistant, University of California, Merced.
- **History of Anthropological Thought and Practice**, Dr. Annith Hundle, Teaching Assistant, University of California, Merced (Fall 2017).
- **Introduction to Socio-cultural Anthropology**, Dr. Robin Delugan, Teaching Assistant, University of California, Merced (Fall 2016).
- **Farsi Literature and Linguistics**, Instructor, Roshan High school, Tehran, Iran (2012–2015).

GRANTS, AWARDS, AND FELLOWSHIPS

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| 2023 | Graduate Dean’s Dissertation Fellowship (\$20,000), UC Merced. |
| 2023 | Cognitive & Information Science Graduate Student Spotlight Award , UC Merced. |
| 2022 | Diversity and Inclusion Awards for the Cognitive Science Society Conference (\$1,000), Cognitive Science Society. |
| 2022 | Grace Hopper Celebration Scholarship to attend GHC conference for Women in Tech, AnitaB.org Foundation. |
| 2021 | Graduate Student Opportunity Program (\$45,000), UC Merced. |
| 2021 | John Templeton Foundation grant (\$30,000), for “A Beginner’s Guide to Neural Mechanism,” a project that provides online and open-access academic videos on topics of neuroscience and philosophy, in collaboration with Fabrizio Calzavarini, Zina Ward, Nick Byrd, and Rafael Gerraty. |
| 2021 | John Templeton Foundation grant (\$8,000), for “No Need to Go It Alone: Ecologically Valid Studies of Group Reasoning,” in collaboration with Daina Carafa, Nick Byrd, Austin Baker, and Trey Boone. |
| 2019 | Fellowship for the Summer Seminars in Neuroscience and Philosophy (SSNAP) (\$1,000), Duke University. |
| 2014 | Graduated with honors as the top student in both Archaeology and Literature and Linguistics, University of Tehran. |
| 2014 | Ranked 1st in the nation in the entrance exam of Archaeology graduate programs. |
| 2010-14 | Granted the annual top student scholarship of the Archaeology program at University of Tehran four years in a row. |
| 2008 | Winner of the gold medal in the National Olympiad of Literature and Linguistics (ranked 1st among 25,000 participants). |

MENTORSHIP

- As part of the Grad-Excel peer mentor program at the University of California, I volunteered to mentor four junior graduate students. I met with my mentees regularly, ensuring they adjusted well to their new environment, providing information about funding opportunities, and assisting them in navigating the challenges they encountered in graduate school.

OUTREACH AND COMMUNITY SUPPORT

- **Data Science for Educational Equity (K–12 Charter School Consortium), San Diego, CA.** Applied computational and machine learning methods to identify at-risk students and support data-informed, personalized learning strategies aimed at reducing achievement gaps.
- **Migrant Education Enrichment in Science and Technology (MESAT), University of California, Merced.** Collaborated in designing and evaluating STEM workshops for K–12 immigrant students; conducted pre/post assessments demonstrating increased scientific understanding and interest in college education.
- **Advocacy for International and Undocumented Students, University of California, Merced.** Developed and disseminated a centralized funding resource identifying internal and external grants accessible to international and undocumented students to support equitable access to research opportunities.
- **Promoting Accessible Neuroscience Education through Free, Online Learning Tools.** Contributed to the development of open-access online videos in neuroscience and philosophy to promote equitable access to foundational educational materials.
- **Society for the Protection of Working and Unhoused Children, Iran.** Volunteered with a nonprofit organization serving working and unhoused children; taught foundational reading and writing skills and helped connect children and their families with educational support services.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Applied AI Microcredential, Georgetown University — 2025**
Completed 8-module instructor-reviewed program on applied generative AI. Developed competencies in prompting, research applications, ethical considerations, privacy, fine-tuning, and critical evaluation of AI outputs.
- **Generative AI Fundamentals, Databricks — 2024**
Gained foundational knowledge of generative AI concepts, responsible AI use, and applications within Databricks' ecosystem.
- **Python for Data Science and Machine Learning Bootcamp, Pierian Training — 2023**
Intensive applied course covering Python, data analysis, visualization, and machine learning techniques using libraries such as NumPy, pandas, scikit-learn, and TensorFlow.
- **Human Resources Analytics, Coursera — 2023**
Coursework in workforce data analysis, predictive modeling, and data-driven HR decision-making.
- **Principles of Pedagogy, University of California, Merced — 2020**
Completed an intensive pedagogy workshop focused on hands-on lesson planning, assignment design, establishing guidelines for productive discussions, and implementing evidence-based teaching strategies.

TECHNICAL SKILLS

Programming Languages

R (Tidyverse, ggplot, rstan, brms), Python (Pandas, Numpy, PyTorch, scikit-learn, Keras, TensorFlow), MATLAB, SQL.

Statistical Analyses

Bayesian statistics, Stochastic modeling, Mixed-effects and logistic regression, Time series analysis.

Machine Learning and AI Techniques

Deep Learning, Natural Language Processing (NLP), Text Embeddings and Vector Databases, Classification, Decision Trees, Random Forest, Clustering, Principal Component Analysis (PCA), Fine-tuning and evaluation of Large Language Models (LLMs), Prompt engineering, AI bias analysis.

Methods

Experimental design, Survey design, Qualtrics, Analysis of naturalistic data, Computational modeling, Human–AI interaction research design.

Tools

Eye-tracking, Virtual reality, EEG, Machine learning and AI frameworks (e.g., Hugging Face, OpenAI API).

Software & Web Development

Software Development, Web Application Development, API Integration, Git/GitHub

Languages

English, Farsi, French.